INCH-POUND
MIL-T-24180A(SH)
15 September 1988
SUPERSEDING
MIL-T-24180(SHIPS)
1 December 1965
(See 6.8)

MILITARY SPECIFICATION

TILE, RUBBER, UNDERWATER ACOUSTIC DECOUPLER

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1 SCOPE

1 1 Scope This specification lists the requirements for rubber tile to be used for reducing the transmission of vibration from metal structures to water, and for reflecting underwater sound

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards The following specifications and standards form a part of this document to the extent specified herein Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

PPP-B-636 - Boxes, Shipping, Fiberboard
PPP-F-320 - Fiberboard. Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MILITARY

MIL-L-19140 - Lumber and Plywood, Fire-Retardant Treated
MIL-P-24441 - Paint, Epoxy-Polyamide General Specification for
MIL-P-24441/1 - Paint, Epoxy-Polyamide, Green Primer, Formula 150, Type I
MIL-P-24441/2 - Paint, Epoxy-Polyamide, Exterior Topcoat, Haze Gray, Formula 151, Type I
MIL-P-24441/3 - Paint, Epoxy-Polyamide, Topcoat, White, Formula 152, Type I.
MIL-A-24456 - Adhesive for Plastic Vibration-Damping Tiles

STANDARDS

MILITARY

MIL-STD-407 - Visual Inspection Guide for Rubber Molded Items
MIL-STD-2073-1 - DoD Materiel Procedures for Development and Application of Packaging Requirements

(Unless otherwise indicated, copies of federal and military specifications and standards are available from the Naval Publications and Forms Center (ATTN NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099)

2 2 Non-Government publications The following documents form a part of this document to the extent specified herein Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6 2)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 412 - Standard Test Methods for Rubber Property in Tension (DoD adopted)
D 413 - Standard Test Methods for Rubber Property - Adhesion to Flexible Substrate (DoD adopted)
D 471 - Standard Test Method for Rubber Property - Effect of Liquids (DoD adopted)
D 573 - Standard Test Method for Rubber - Deterioration in an Air Oven (DoD adopted)
D 945 - Standard Test Methods for Rubber Properties in Compression or Shear (Mechanical Oscillograph). (DoD adopted)
D 2240 - Standard Test Method for Rubber Property - Durometer Hardness (DoD adopted)
D 3951 - Standard Practice for Commercial Packaging (DoD adopted)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents These documents also may be available in or through libraries or other informational services)

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2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6 4) in accordance with 4.3.

3 2 Materials The rubber shall be polymerized chloroprene compounded and vulcanized to have the specified properties A suggested formulation is given in 6 5

3 2 1 Recovered materials Unless otherwise specified herein, all material incorporated in the products covered by this specification shall be new and may be fabricated using materials produced from recovered materials to the maximum extent practicable without jeopardizing the intended use The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification unless otherwise specifically specified

3 3 Construction The tile shall have the construction and dimensions shown on figure 1 Tiles other than 12 by 12 inches shall be as specified (see 6 2) The face of the base sheet and all four edges of the tile shall be roughened by buffing or sanding so that no further processing is required for adhesion to steel except solvent cleaning and cement application The other (outer) face (cover sheet) of the tile shall be marked in colored ink with the manufacturer's name or trademark, the month and year of manufacture, contract number and specification number

3 4 Weight The tile shall weigh 2 8 $\pm$ 0.2 pounds

3 5 Airtightness The tile shall show no blisters or indication of leakage (see 4.5.1).

3.6 Adhesion of components A load of 20 pounds shall not separate at a rate greater than 1 inch per minute the cover from the base tile or the tile from the preserved steel surface before aging, after oven aging, and after water immersion (see 4 6.1)

3 7 Properties of rubber. The rubber in the tile shall conform to the requirements specified in table I.

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TABLE I Physical requirements of rubber

Physical properties	Maximum	Minimum	Test
Mechanical properties under compression			4 6 2
Static modulus, lb/in^2	1,030	---	
Dynamic modulus, lb/in^2	1,620	---	
Resilience, percent	---	70	
Tensile strength, lb/in^2	---	2,500	4 6 3
Decrease after oven aging, percent	20	---	4 6 3 1
Ultimate elongation, percent	---	450	4 6 3
Decrease after oven aging, percent	20	---	4 6 3 1
Durometer hardness	65	55	4 6 4
Volume swell after water immersion, percent	6	0	4 6 5

3 8 Workmanship The finished item shall be clean, well made and free from defects which may affect the serviceability

4 QUALITY ASSURANCE PROVISIONS

4 1 Responsibility for inspection Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements

4 1 1 Responsibility for compliance All items shall meet all requirements of sections 3 and 5 The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract Sampling inspection, as part of the manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material

4 2 Classification of inspections The inspection requirements specified herein are classified as follows·

(a) First article inspection (see 4 3).
(b) Quality conformance inspection (see 4 4)

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4 3 First article inspection First article inspection shall consist of the examination of 4 5 1 and tests of 4 6 The inspection material shall consist of a sample identical in composition and equivalent in cure to the material to be provided for quality conformance inspection (see 6 3) The first article sample shall conform to the requirements specified in this specification First article tests shall be monitored by a Government representative. The contractor shall not proceed with production without receiving written approval from the contracting activity (see 6.3)

4 4 Quality conformance inspection. Quality conformance inspection shall consist of the examinations and tests specified in 4 5

4 4 1 Sampling for quality conformance

4 4 1 1 Lot. Tiles of the same dimensions and made of the same compound and offered for delivery at one time shall be considered a lot for purposes of sampling and inspection In each case the number of pieces shall be the lot size

4 4 1 2 Sampling for visual, dimensional and airtightness examinations Sample tiles shall be taken at random from each lot in accordance with the sampling plan in table II for the examinations specified in 4 5 1

TABLE II Sampling for visual, dimensional and airtightness examinations

Lot size	Sample size	Number of nonconforming or defective rubber tiles			
		Major defects		Total defects (major plus minus)	
Number of rubber tiles	Number of rubber tiles	Acceptance number	Rejection number	Acceptance number	Rejection number
1 to 4	All	0	1	1	2
5 to 9	5	0	1	1	2
10 to 25	8	0	1	1	2
26 to 62	13	0	1	1	2
63 to 160	20	1	2	2	3
161 to 410	32	1	2	3	4
411 to 1,000	50	2	3	4	5
1,001 to 2,600	80	3	4	6	7

4 4 1.2 1 A major defect is a defect that is likely to result in failure, or to reduce materially the usability of the unit of product for its intended purpose

4 4 1 3 Sampling for destructive tests Sample tiles shall be taken at random from those selected in 4 4 1 2 in sufficient quantity to conduct the tests specified in 4.5.2.1 or 4 5 3 1, as applicable

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4.4 1 3 1 Additional rubber samples for test Two sheets, each 6 by 6 by 0 080 ± 0 010 inch thick and one slab 4 by 4 by 0 500 inch thick shall be furnished with each lot The tests specified in 4 5.2 2 and 4.5 3.2, as applicable, shall be conducted on these rubber samples

4 5 Quality conformance examinations and tests

4 5 1 Visual, dimensional, and airtightness examinations Each of the samples taken in accordance with 4 4.1 2 shall be subjected to an examination for workmanship, dimensions, tolerances and airtightness MIL-STD-407 shall be used to determine and evaluate defects through visual examination The examination for airtightness shall be conducted as follows

(a) Place a single layer of tiles on the platen of a press or between hot plates maintained at 152 9 degrees Celsius (°C) (310 degrees Fahrenheit (°F)), and arranged to close as specified below
(b) Close the press or the hot plates against the tiles at the lowest pressure which can be maintained against 1/2-inch thick spacer bars
(c) Two minutes after placing the tiles on the hot surface, release the pressure and remove the heated tiles as rapidly as possible
(d) Examine each heated tile by picking it up and tilting to cross light the patterned surface Tile which is airtight will exhibit a symmetrical pattern of raised areas over each hole Areas which are not raised over the holes indicate leakage Blisters which destroy the symmetry indicate faulty bonding within the tile Any tile exhibiting such blisters or indication of leakage shall be rejected and classified as a major defect

4 5 1 1 Rejection Any defective tile shall not be offered for delivery If the number of defective tiles in any sample exceeds the applicable acceptance number shown in table II, this shall be cause for rejection of the lot represented by the sample

4.5.2 Quality conformance tests

4 5 2 1 Tests on the tile Each sample taken in accordance with 4 4 1 3 shall be subjected to the following tests·

Test	Test procedure
Adhesion, initial	
Cover to base of tile	4 6.1 1
Tile to steel	4.6 1.2

4 5 2.2 Tests on rubber samples Each sample selected in accordance with 4 4.1.3.1 shall be subjected to the following tests

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Test	Test procedure
Static modulus, dynamic modulus and resilience	4.6.2
Tensile strength, initial	4.6.3
Ultimate elongation, initial	4.6 3
Hardness, Shore A durometer	4 6 4

4 5.2 3 Rejection for nonconformance. If any of the samples is found not to conform with the requirements of this specification, this shall be cause for rejection of the lot which it represents.

4.5 3 Production check tests Production check tests shall be conducted on the first lot of material offered for delivery under a contract or order and on every fifth lot thereafter

4 5 3 1 Tests on the tile. Each sample taken in accordance with 4.4.1.3 and shall be subjected to the following tests.

Test	Test procedure
Adhesion, after oven aging	
Cover to base of tile	4 6 1 1.1
Tile to steel	4.6 1 2.1

4.5 3 2 Tests on rubber samples Each sample selected in accordance with 4.4 1.3.1 shall be subjected to the following tests

Test	Test procedure
Tensile strength, after oven aging	4.6 3.1
Ultimate elongation, after oven aging	4.6 3.1

4 5 3 3 Rejection If any production check test sample is found to be not in conformance with this specification, this shall be cause for rejection of the lot which it represents Furthermore, additional samples shall be selected from (or to represent) each subsequent lot and shall be subjected to the test or tests wherein failure occurred A lot shall then be accepted only upon satisfactory completion of the test or tests by all the samples selected to represent the lot This additional testing shall be discontinued when four successive lots have passed the test or tests

4 6 Test procedures Unless otherwise indicated in the particular test procedure, tests shall not be conducted prior to conditioning the test specimens in air for 4 hours at 27 ± 5°C (80 ± 9°F) Sample preparation may be undertaken without regard to this time interval

4 6 1 Adhesion. The adhesion of the cover to the base of the tile and the tile to steel shall be determined initially, after oven aging, and after water immersion

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4 6 1 1 Adhesion of cover to base of tile The test for adhesion of the cover to the base of the tile shall be determined in accordance with ASTM D 413, static-mass, method, 180-degree peel, using type A strip specimens

4.6 1 1 1 Oven aging Two ASTM D 413, type A, 180-degree peel, strip specimens shall be subjected to accelerated aging in an oven in accordance with ASTM D 573 The time shall be 166 ± 1 hours, the temperature shall be 70 ± 1 1°C (158 ± 2°F) After oven aging, the test described in 4 6 1 1 shall be performed

4 6 1 1 2 Water immersion Two ASTM D 413, type A, 180-degree peel, strip specimens shall be subjected to immersion in distilled water The temperature of the water shall be 54 3 ± 1 1°C (130 ± 2°F), and the pressure shall be 600 ± 10 pounds per square inch gauge The duration of the immersion under these conditions shall be 166 ± 1 hours At the end of this period, the specimens shall be removed from the warm water and immersed in open distilled water at room temperature for 1 to 6 hours The test procedure specified in 4 6 1 1 shall be started within 3 minutes after removal from the latter water

4 6 1 2 Adhesion of tile to steel The test for adhesion of the tile to a preserved steel surface shall be performed by the following procedure

(a) Two strip specimens, each 1-1/2 inches wide and not less than 8 inches long, shall be cut from the tile at a 45-degree angle to the sides
(b) The buffed surface of each strip shall be scrubbed with a rag wet with 1-1-1 trichloroethane (methyl chloroform)
(c) A rigid steel plate having a surface area larger than the strip specimen shall be sandblasted to white metal The sandblasted area shall be brushed to remove sand particles and wiped with a clean lintless rag which has been wet with the solvent described in (b) The sandblasted surface shall then be preserved with one coat each of the following Formula 150 green epoxy-polyamide primer in accordance with MIL-P-24441 and MIL-P-24441/1, formula 151 haze gray epoxy exterior topcoat in accordance with MIL-P-24441 and MIL-P-24441/2, formula 152 white epoxy topcoat in accordance with MIL-P-24441 and MIL-P-24441/3 Each coat of the preservative shall be allowed to cure not less than 24 hours before applying the next coat The final coat shall be allowed to cure not less than 48 hours Before cementing the specimens to the steel plate, the final coat of paint shall be carefully hand sanded with 60 grit abrasive cloth and then cleaned with a lint-free cloth wetted with 1-1-1 trichloroethane
(d) One coat of an adhesive conforming to the requirements of MIL-A-24456 shall be trowelled onto the buffed and clean rubber surface The trowel shall consist of sheet metal provided with equilateral triangular serrations of 8 teeth to the inch
(e) The cemented surface of each rubber strip shall be rolled down against the clean preserved steel surface, using a steel hand roller to press the rubber firmly against the steel at all points The assemblies shall be allowed to stand for 4 days, but not more than 7 days, at 70 to 80°F before testing for adhesion

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(f) The assemblies shall be tested for adhesion with a 90-degree angle of pull The metal-strip assembly shall be supported in a horizontal position with the strip on the underside A sufficient length of the strip shall be separated from the steel to permit the attachment of a clamp. A 20-pound dead-weight load (including clamp) shall be gently suspended from the loose end of the strip, and the rate of separation, if any, shall be recorded

4 6 1 2 1 Oven aging Two metal-strip assemblies prepared in accordance with 4 6 1 2 shall be subjected to accelerated aging in an oven in accordance with 4 6 1 1.1. After removal from the oven, the test described in 4 6 1 2(f) shall be performed within 16 to 96 hours

4 6 1 2.2 Water immersion. Two metal-strip assemblies prepared in accordance with 4.6.1 2 shall be subjected to water immersion in accordance with 4 6 1 1 2 After water immersion, the test described in 4 6 1 2(f) shall be performed within 16 to 96 hours.

4 6 2 Mechanical properties under compression The static modulus, dynamic modulus and resilience of the tile shall be determined using a Yerzley oscillograph following the procedure and method of calculation specified in ASTM D 945 Test specimens shall be cylinders 0 750 ± 0 005 inch in diameter and 0 50 ± 0 01 inch thick died from the 0 5-inch slab provided in accordance with 4 4 1 3 1

4.6 3 Tensile strength and ultimate elongation The tensile strength and ultimate elongation of the tile shall be determined in accordance with ASTM D 412 using dumbbell specimens cut with die C from the sheets provided in accordance with 4 4 1 3 1

4 6 3 1 Tensile specimens Accelerated aging of the tensile specimens shall be performed in accordance with ASTM D 573.

4 6 4 Durometer hardness. The hardness of the 0 5-inch slab provided in accordance with 4.4.1.3 1 shall be measured with a Shore A durometer as specified in ASTM D 2240 A 3-second reading shall be taken

4 6 5 Volume change in water Specimens shall be 1 by 2 inches by 0 08 to 0 09 inch They shall be prepared from the sheets provided in accordance with 4.4 1 3 1 The change in volume of these specimens after 166 ± 1 hours in distilled water at 54 3 ± 1.1°C (130 ± 2°F) shall be determined in accordance with ASTM D 471

4.7 Inspection of packaging Sample packages and packs, and the inspection of the preservation, packing and marking for shipment, stowage and storage shall be in accordance with the requirements of section 5 and the documents specified therein

5 PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition.)

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5 1 Preservation Preservation shall be level A, C, or commercial as specified (see 6 2)

5.1.1 Level A Fifteen tiles shall be unit packed flat in a snug-fitting fiberboard box conforming to PPP-B-636, class-weather-resistant with the type, variety, grade and box style at the contractor's option A reinforcing liner and top and bottom pads of the same material as the fiberboard box shall be used Tiles shall be separated from one another and the pads by a minimum of 4 mil (0 10 millimeter) thick polyethylene film Separate pieces of polyethylene film shall be used between each tile and each separator shall be 13 inches (33 centimeters) long by 13 inches wide Box closure shall be in accordance with 5 2 2 1 2

5 1 2 Level C Preservation shall be as specified for level A, except that the fiberboard box shall be class-domestic and in accordance with 5 2 1 1 1 2 and 5 2 2 1.

5 1 3 Commercial Preservation shall be in accordance with ASTM D 3951

5 2 Packing Packing shall be level A, B, C, or commercial as specified (see 6 2).

5 2 1 General requirements for levels A, B, and C

5 2 1 1 Containers Containers selected (see 5 2 2) shall be of minimum weight and cube to be consistent with the protection required, of uniform size, and contain identical quantities of identical tiles.

5 2.1 1 1 Navy fire-retardant requirements

5 2.1 1.1 1 Lumber and plywood Unless otherwise specified (see 6 2), all lumber and plywood including laminated veneer material used in shipping container and pallet construction, members, blocking, bracing, and reinforcing shall be fire-retardant treated material conforming to MIL-L-19140 as follows:

Levels A and B - Type II - weather resistant
Category 1 - general use
Level C - Type I - non-weather resistant
Category 1 - general use

5 2 1 1.1 2 Fiberboard Fiberboard used in the construction of class-domestic, non-weather resistant fiberboard and cleated fiberboard boxes shall meet the flammability and smoke requirements of PPP-F-320 and amendments thereto

5 2.2 Levels A, B, and C containers Tiles preserved as specified (see 5.1) shall be packed in shipping containers The level of packing (see 6.2) shall be as stated in MIL-STD-2073-1, appendix C, and herein Unless otherwise specified (see 6 2), container selection shall be at the contractor's option

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5 2 2 1 Caseliners, closure, and gross weight.

5 2 2 1 1 Caseliners Unless otherwise specified (see 6 2), level A shipping containers shall be provided with waterproof caseliners in accordance with MIL-STD-2073-1

5 2 2 1 2 Closure Container closure, reinforcing, or banding shall be in accordance with the applicable container specification or appendix thereto, except that weather-resistant fiberboard boxes shall be closed in accordance with method V and reinforced with non-metallic or tape banding and domestic fiberboard boxes shall be closed in accordance with method I using pressure-sensitive tape

5.2 2 1.3 Weight Wood, plywood, and cleated type containers exceeding 200 pounds gross weight shall be modified by the addition of skids in accordance with MIL-STD-2073-1 and the applicable container specification or appendix thereto

5 2 3 Commercial Tiles preserved as specified (see 5 1) shall be packed for shipment in accordance with ASTM D 3951 and herein

5 2 3 1 Container modification Shipping containers exceeding 200 pounds gross weight shall be provided with a minimum of two, 3- by 4-inch nominal wood skids laid flat, or a skid- or sill-type base which will support the material and facilitate handling by mechanical handling equipment during shipment, storage, and stowage

5 3 Palletized unit loads When specified (see 6 2), shipping containers shall be palletized in accordance with MIL-STD-2073-1

5 4 Marking

5 4.1 Levels A, B, and C In addition to any special marking required (see 6 2 and herein), unit packs, shipping containers, and palletized unit loads shall be marked in accordance with MIL-STD-2073-1 The marking shall include lot numbers and bar coding (unit packs and shipping containers) In addition, each unit and shipping container shall be marked on two sides with the word "UP" An arrow pointing toward the top of the container shall be placed beneath each word Arrows shall be employed only to indicate or supplement the words, "This side UP". The following precautionary markings shall appear on two opposite sides of each package or packing

"FOR STORAGE DO NOT STACK MORE THAN TWO HIGH
STORE IN COOL DRY PLACE

FREE OF ASBESTOS AND CERAMIC
(REFRACTORY) FIBERS."

5.4 2 Commercial Marking shall conform to 5.4 1, except that ASTM D 3951 applies

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6 NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6 1 Intended use The rubber tile is intended to be used for reducing the transmission of vibration from metal structures to water, and for reflecting underwater sound

6 2 Acquisition documents Acquisition documents must specify the following

(a) Title, number, and date of this specification
(b) Issue of DoDISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced (see 2 1 1 and 2 2)
(c) When first article is required (see 3 1)
(d) Size of sheet, if other than 1 foot by 1 foot tile (see 3 3)
(e) When container selection is not at contractor's option (see 5 2 2)
(f) Level of preservation and level of packing required (see 5 1 and 5 2).
(g) When Navy fire-retardant requirement for lumber and plywood is not required (see 5 2 1 1 1 1)
(h) When caseliners are not required (see 5 2 2 1 1)
(i) Palletization when required (see 5 3)
(j) Special markings required (see 5 4 1)

6 3 Consideration of data requirements The following data requirements should be considered when this specification is applied on a contract The applicable Data Item Descriptions (DIDs) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/ provided and that the DIDs are tailored to reflect the requirements of the specific acquisition To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DoD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423

Reference paragraph	DID number	DID title	Suggested tailoring
4.3, 4 4 1.3 1	DI-E-2121	Certificate of compliance	--------
4 3	DI-T-4902	First article test report	--------

The above DIDs were those cleared as of the date of this specification The current issue of DoD 5010 12-L, Acquisition Management Systems and Data Requirements List (AMSDL), must be researched to ensure that only current, cleared DIDs are cited on the DD Form 1423.

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6 4 First article When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first _____ production items, a standard production item from the contractor's current inventory (see 3 1), and the number of items to be tested as specified in 4 3. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract Bidders should not submit alternate bids unless specifically requested to do so in the solicitation

6 5 The material formulation shown in table III is offered only as a suggestion The Government does not guarantee that tiles made from this formulation will conform to the requirements of this specification

TABLE III Suggested formulations for rubber stock

Ingredients	Parts by weight
Neoprene type GRT	100
Stearic acid	1
Phenyl-alpha-napthylamine	2
SRF carbon black	40
Red lead	15

6 6 Data content Certification (see 4.3 and 4.4 1 3 1) should indicate the following.

(a) That material submitted for first article inspection is identical in composition and equivalent in cure to the material that will be provided for quality conformance inspection
(b) That additional test samples (see 4.4.1.3.1) are identical in cure and composition and prepared at the same time as material used in the submitted lot.

6.7 Subject term (key word) listing.

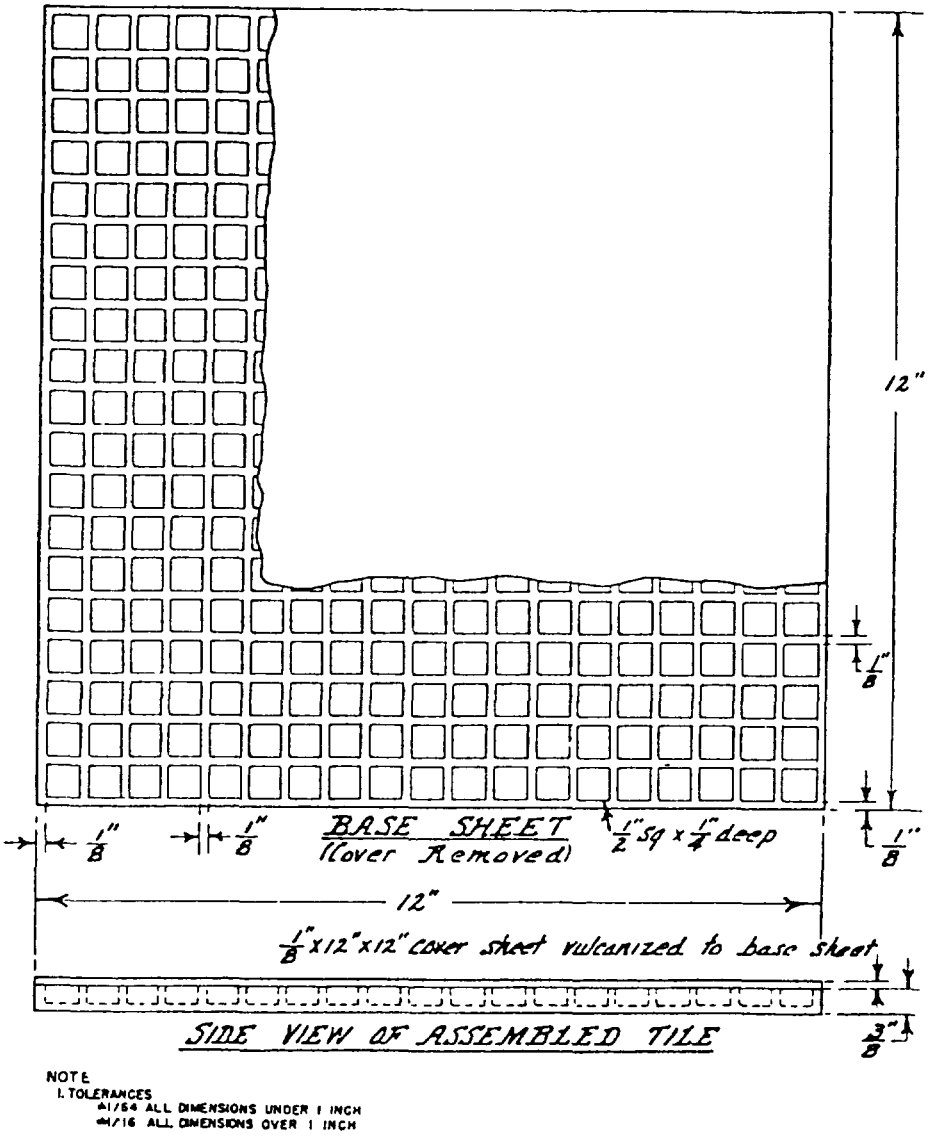
Airtight
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Resilience
Static modulus

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6 8 Changes from previous issue Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes

Preparing activity
Navy - SH
(Project 9320-N006)

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SH 13231889

FIGURE 1 Construction of rubber tile for vibration decoupling

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NOTE This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions - Reverse Side)

1 DOCUMENT NUMBER	2 DOCUMENT TITLE
MIL-T-24180A(SH)	TILE, RUBBER, UNDERWATER ACOUSTIC DECOUPLER

3a. NAME OF SUBMITTING ORGANIZATION	4 TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)* _____
b ADDRESS *(Street City State ZIP Code)*	

5 PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6 REMARKS

7a NAME OF SUBMITTER *(Last First, MI)* – Optional	b WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c MAILING ADDRESS *(Street, City State ZIP Code)* – Optional	8 DATE OF SUBMISSION *(YYMMDD)*

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